



Introduction to Machine vision & Motion analysis

CSE0613409

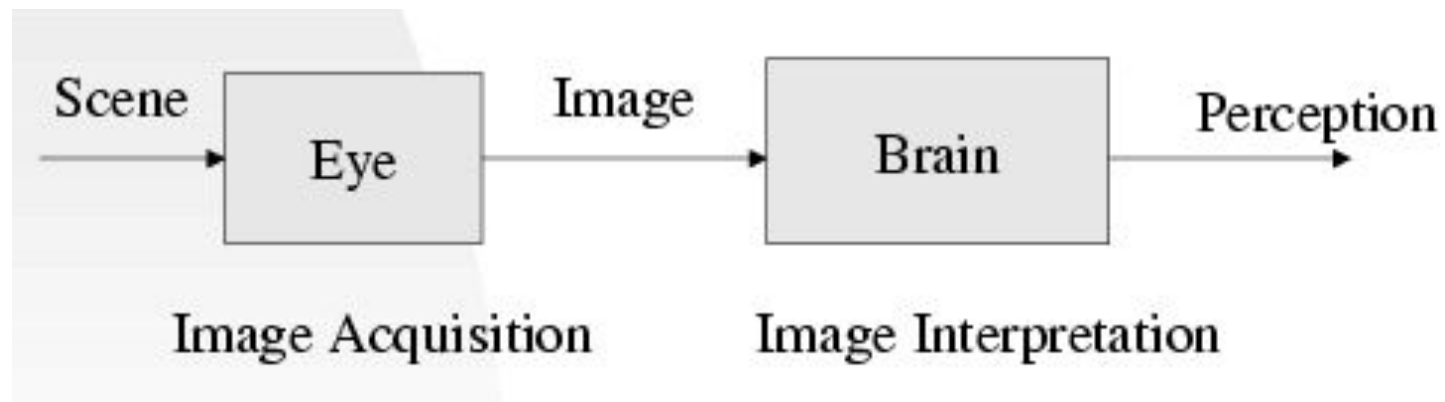
Machine Vision and Motion Analysis

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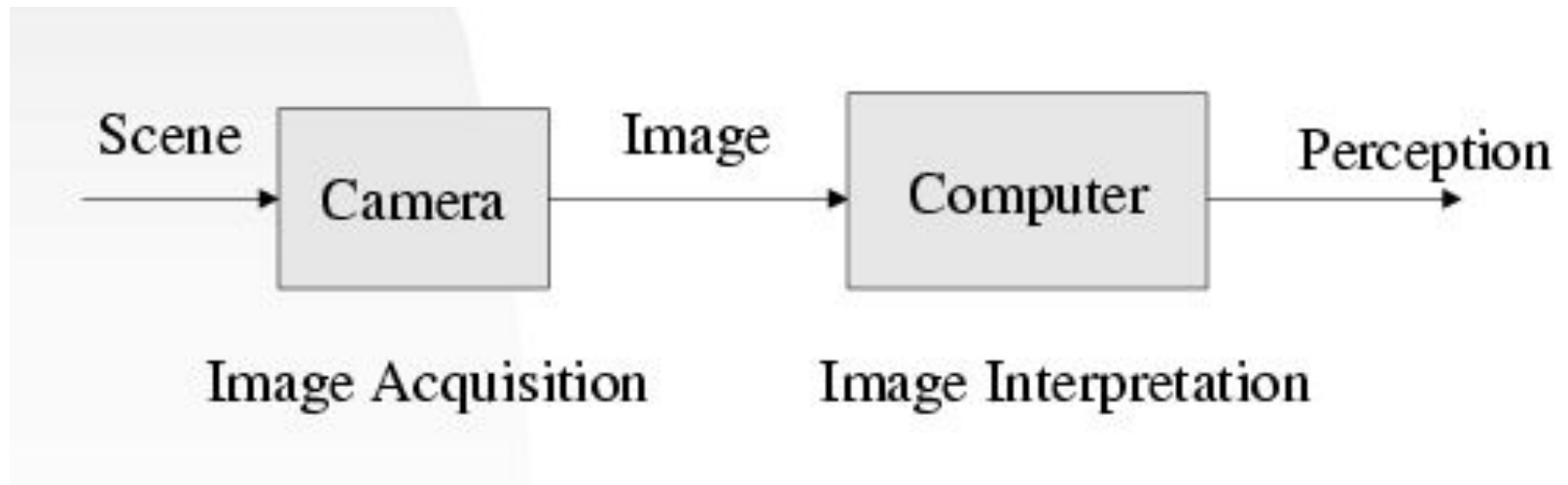
Human Vision

- Vision is the process of discovering what is present in the world and where it is by looking.



Computer Vision

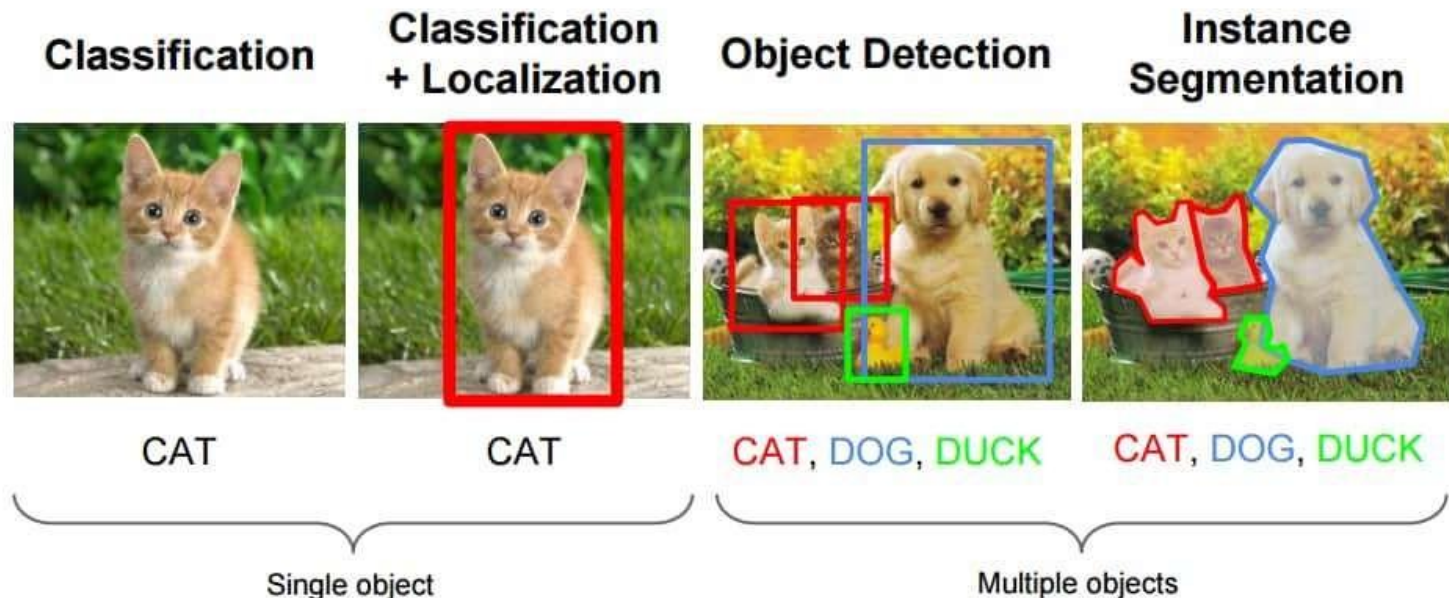
- Computer Vision is the study of analysis of pictures and videos in order to achieve results similar to those as by men.



What is meant by computer vision?

- **Computer vision** tasks include methods for acquiring, processing, analyzing and understanding digital images, and extraction of high-dimensional data from the real world in order to produce numerical or symbolic information, e.g., in the forms of decisions.

Computer Vision Tasks

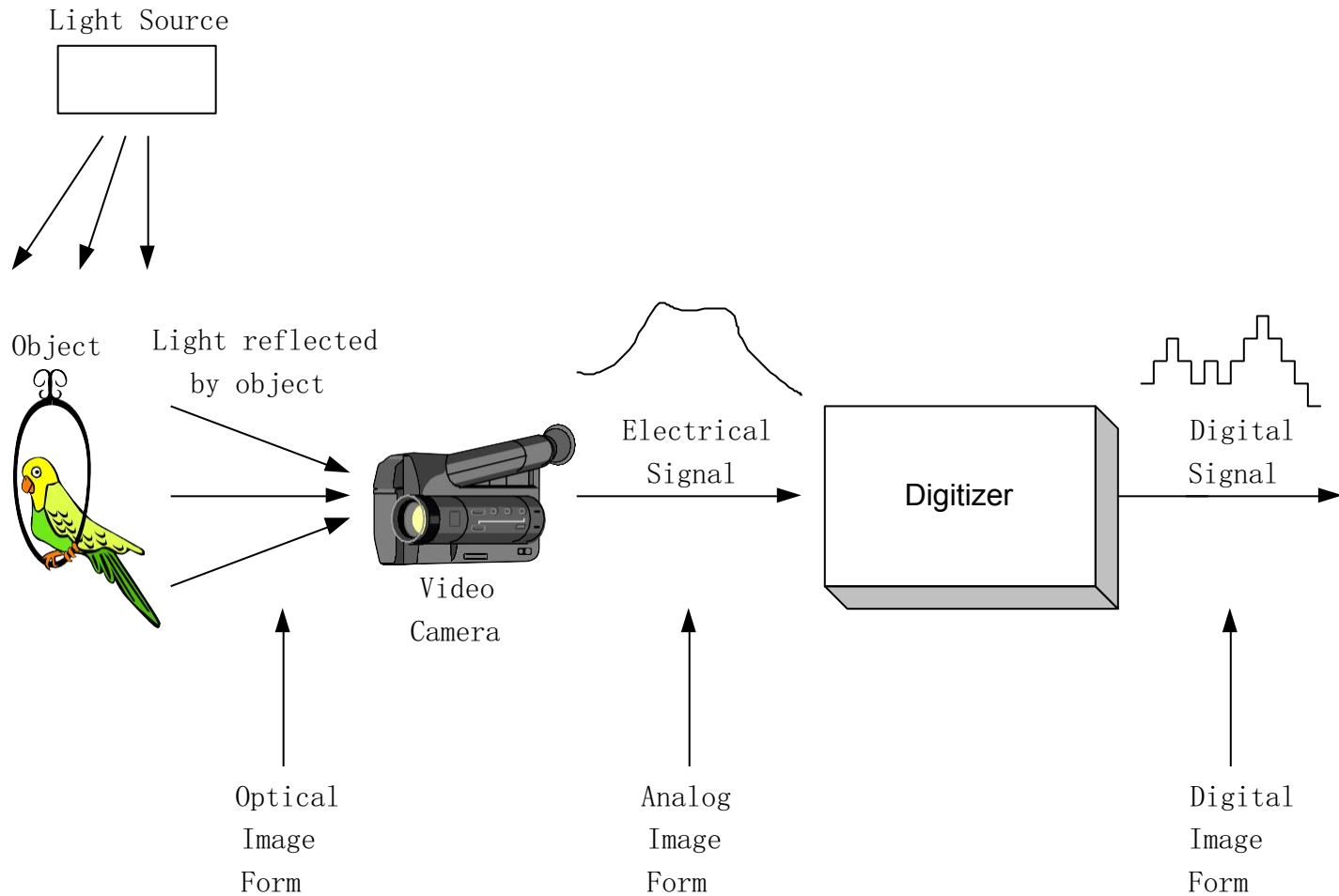


What is computer vision used for?

- **Computer vision** is a field of artificial intelligence that trains computers to interpret and understand the **visual** world. Using digital images from cameras and videos and deep learning models, machines can accurately identify and classify objects — and then react to what they “see.”



Acquisition of digital image



What is image processing and computer vision?

- **Image processing** is a subset of **computer vision**.
- A **computer vision** system uses the **image processing** algorithms to try and perform emulation of **vision** at human scale.
- For example, if the goal is to enhance the **image** for later use, then this may be called **image processing**.

Types of computer vision

- Lower-level vision
 - Analyzing textures, edges and gradients in images, without concern for the semantics (e.g. objects) of the image
- Higher-level vision
 - Making predictions about the semantics or higher-level functions of content in images (e.g. objects, attributes, styles, motion, etc.)
 - Involves machine learning

What is computer vision?

- Automatic understanding of images and video
 - Algorithms and representations to allow a machine to recognize objects, people, scenes, and activities (*perception and interpretation*)
 - Algorithms to mine, search, and interact with visual data (*search and organization*)
 - Computing properties of the 3D world from visual data (*measurement*)

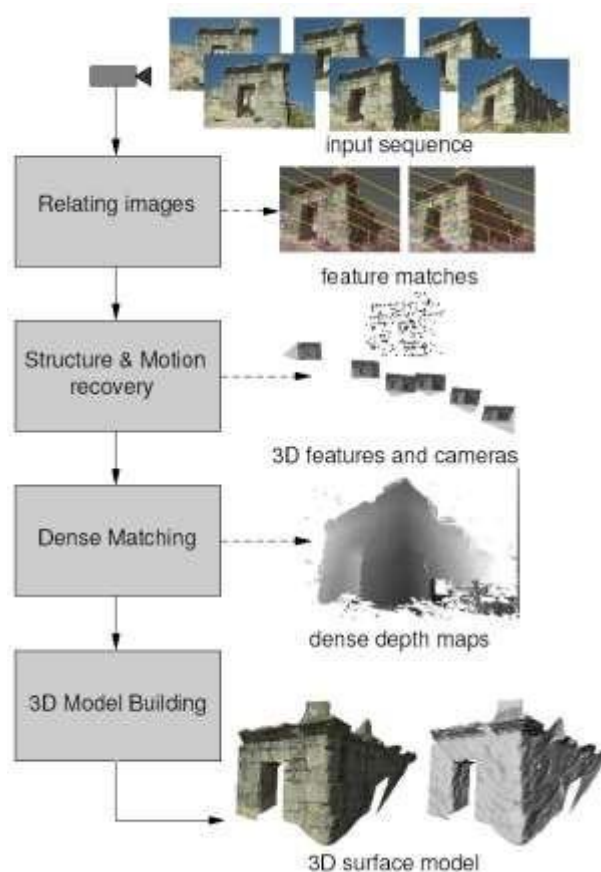
Vision for measurement

Real-time stereo



Pollefeys et al.

Structure from motion



Multi-view stereo for community photo collections



Goesele et al.

Why vision?

- As image sources multiply, so do applications
 - Relieve humans of boring, easy tasks
 - Human-computer interaction
 - Perception for robotics / autonomous agents
 - Organize and give access to visual content
 - Description of image content for the visually impaired
 - Fun applications (e.g. transfer art styles to my photos)